

Final Report

Multi-Level Revenue Forecasting and Demand Analysis for Resort Operations

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Tools Used: Python (Jupyter Notebook), SQL, Excel

Executive Summary

This project develops a multi-level revenue forecasting system for resort operations across hotel, ski, and golf departments using time-series analysis. The objective is to improve demand planning and operational decision-making by identifying seasonal and short-term revenue patterns.

Multiple forecasting models, including Moving Average, Exponential Smoothing, ARIMA, and Regression were applied and evaluated. The results reveal strong seasonal behavior in revenue, with ski operations generating peak winter demand, hotel revenue remaining relatively stable throughout the year, and golf revenue showing higher volatility.

Among all models, the Regression model achieved the best forecasting performance with the lowest error values. ARIMA captured revenue trends reasonably well, while Moving Average and Exponential Smoothing served as baseline forecasting models with comparatively higher prediction errors.

The findings support practical business strategies such as seasonal staffing optimization, demand-based pricing during peak periods, and improved resource allocation across resort departments.

Introduction:

Unlike simulation-based forecasting studies, this project applies predictive modelling techniques to real operational business data, increasing the practical relevance and applicability of the forecasting results.

Business problem:

Resort revenue is highly seasonal across hotel, ski, and golf operations, leading to unstable demand and planning challenges. Current forecasting relies on simple historical trends, resulting in inefficiencies in staffing, pricing, and resource allocation.

A structured multi-level forecasting system is required to support better revenue planning and operational decision-making across business units.

Data Overview:

Resort operations follow seasonal business cycles, with certain departments operating only during specific periods of the year. Ski operations are concentrated during the winter season, while golf operations are more active during warmer months. Temporary operational closures and off-season reductions contributed to fluctuations and low-revenue periods observed in the dataset.

The dataset used in this study is internal resort operational sales data for the year 2025, covering hotel, ski, and golf departments.

The data is recorded at a **daily level by department**, and is further aggregated into a **monthly level** for trend and seasonal analysis.

Key variables include:

- Date (daily time series)
- Department (hotel, ski, golf)
- Daily revenue
- Monthly aggregated revenue (derived from daily data)

The dataset enables multi-level analysis of revenue patterns across departments and time horizons.

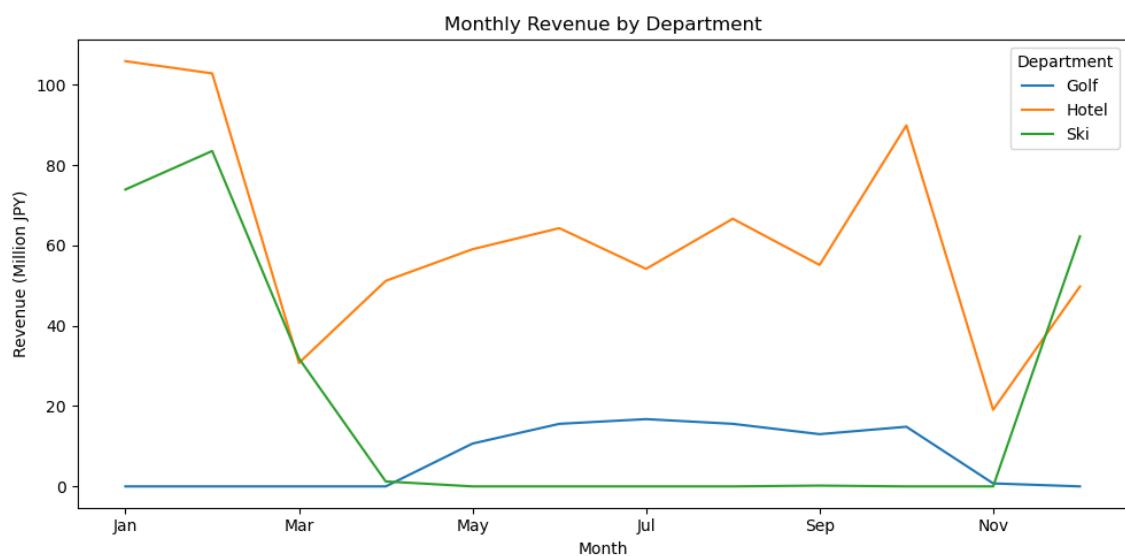
Exploratory Data Analysis:

Exploratory Data Analysis was conducted to understand revenue patterns across hotel, ski, and golf departments and to identify seasonal and structural trends in the dataset.

The dataset includes daily revenue data for the year 2025, which was analyzed at both daily and aggregated monthly levels to capture short-term fluctuations and broader seasonal patterns.

a. Revenue Distribution by Department:

Daily revenue was first analyzed by department to understand the contribution of each business unit.

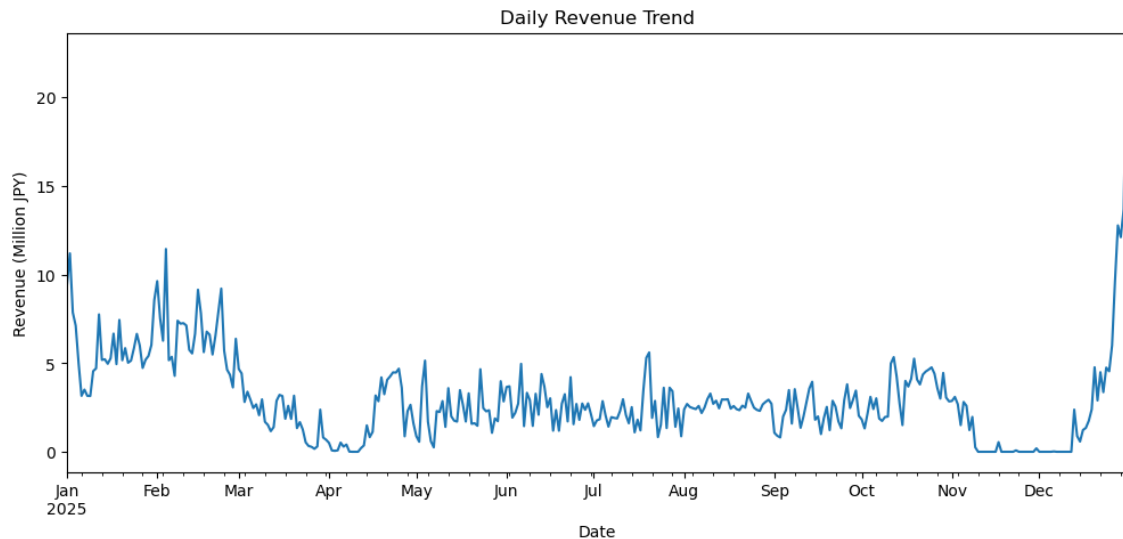


Hotel revenue is the highest and relatively stable, with peaks in winter and autumn months. Ski revenue is highly seasonal, concentrated in winter and almost zero in off-season. Golf revenue is lower and mainly active in mid-year months.

Overall, the resort shows strong seasonality, driven mainly by ski in winter and hotel as the stable revenue source.

b. Time Series Trend Analysis

Revenue trends were analyzed over time to identify patterns and seasonality.



The time series shows a clear seasonal pattern with strong fluctuations in revenue over the year.

Revenue is highest at the beginning of the year (Jan–Feb), followed by a sharp decline through March and April, where it reaches its lowest point. From May to September, revenue remains relatively stable at a low-to-moderate level with minor daily variations, indicating a steady off-peak period.

A slight recovery appears around October, but this is temporary. November shows another decline, dropping close to zero in several periods. The most significant feature is the sharp spike at the end of December, suggesting a strong year-end demand surge.

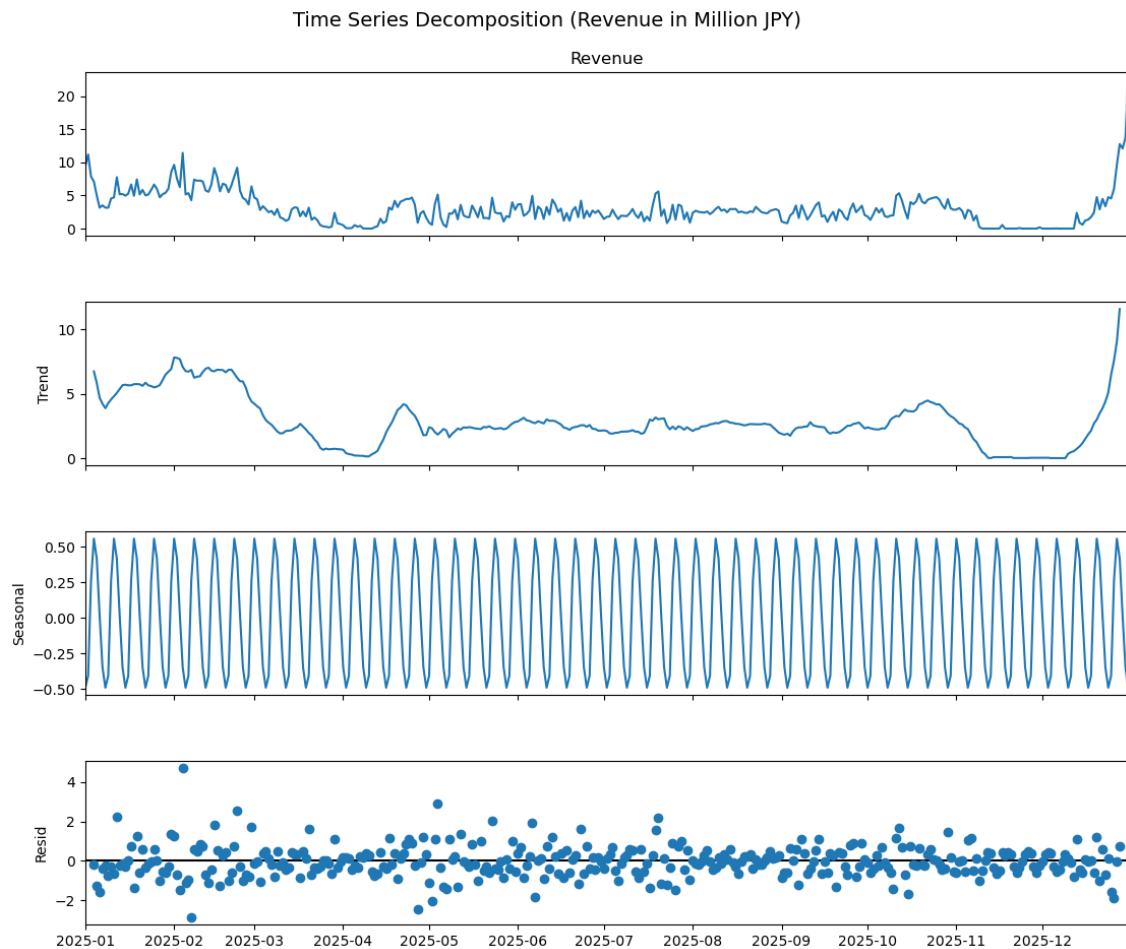
Overall, the trend reflects strong seasonality with peak demand in winter periods and weak performance during the spring and early winter months.

c. Key Observations:

- Ski revenue exhibits high seasonal sensitivity
- Hotel revenue provides stable year-round income
- Golf revenue shows moderate seasonal variation
- Overall revenue patterns confirm the presence of seasonality and demand variability across departments

Time Series Decomposition:

Time series decomposition was applied to separate revenue into trend, seasonal, and residual components to better understand the underlying structure of demand.



The revenue series was decomposed into observed, trend, seasonal, and residual components to better understand underlying demand behavior.

- The **trend component** shows a gradual movement in revenue over time, reflecting structural changes in business performance.
- The **seasonal component** captures strong and consistent cyclical patterns, indicating recurring demand fluctuations across the year.
- The **residual component** represents random noise after removing trend and seasonality, which remains relatively stable and centered around zero.

Overall, the decomposition highlights that revenue is primarily driven by seasonal effects and long-term trend behavior, with limited unexplained variation.

Feature Engineering:

To improve forecasting performance, several time-based features were created to capture demand patterns and external effects:

- Lag variable (Lag_1): captures short-term dependency from previous day revenue
- 7-day rolling average: smooths volatility and highlights underlying trend
- Holiday flag: captures demand shifts during special calendar events
- Calendar features (month, day of week, weekend): capture seasonal and weekly behavior patterns

Modeling Approach:

Multiple forecasting techniques were used to ensure robust revenue prediction by capturing different patterns such as trend, seasonality, and short-term fluctuations.

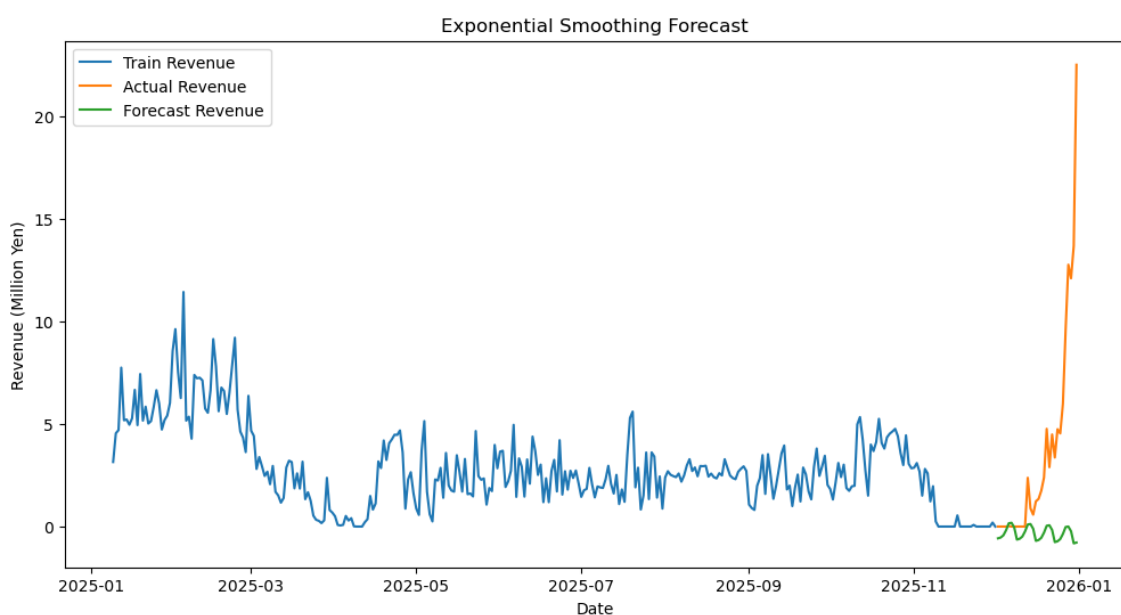
Baseline models

1. Moving Average Baseline Forecast

A simple moving average model was used as a baseline to smooth short-term fluctuations in revenue. This model provides a reference benchmark to evaluate the performance of more advanced forecasting approaches such as ARIMA and SARIMA. Although the detailed plot is not included, it served as a foundational comparison model.

2. Exponential Smoothing forecast

Exponential Smoothing was used to forecast revenue by giving more weight to recent observations to capture short-term trends and fluctuations.



The Exponential Smoothing Forecast captures short-term revenue fluctuations but shows clear divergence from actual revenue in Q4 2025. While it smooths historical patterns effectively, it underestimates the sharp year-end demand surge.

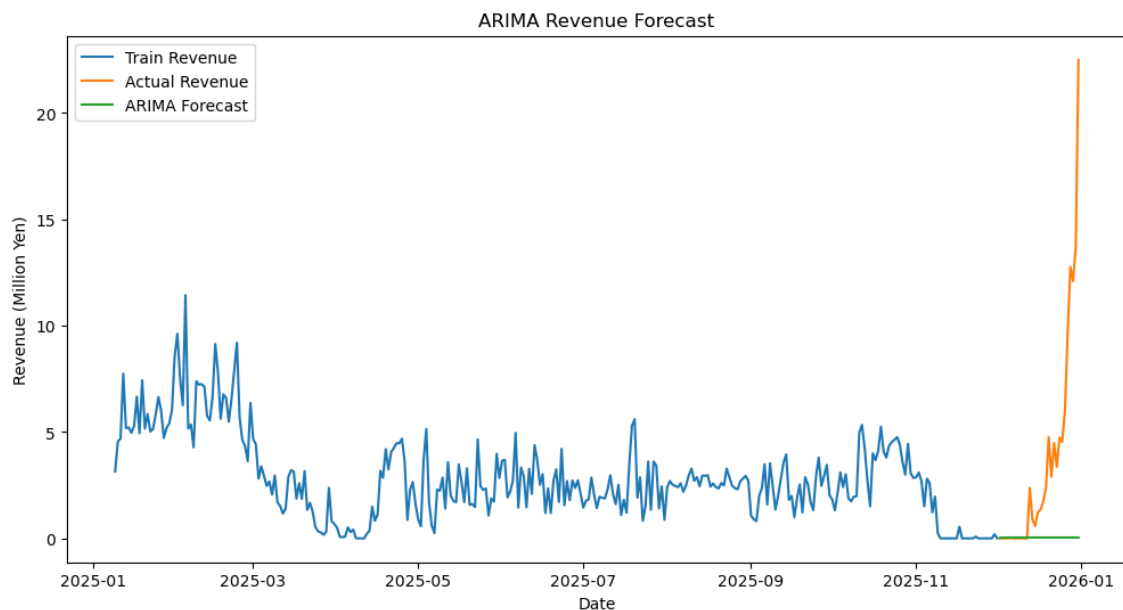
Key points:

- **Model sensitivity:** Reflects minor cycles but is influenced by recent low revenue levels.
- **Trend lag:** Fails to capture sudden demand spike, leading to under-forecasting.
- **Usefulness:** Still effective for short-term planning, but weak for structural changes.

Time series models

3. ARIMA Revenue Forecast:

The ARIMA model was used to forecast revenue by capturing trend and autocorrelation in historical data. It provided a more accurate and stable forecast compared to simple baseline methods, making it a key model for time-series prediction in this analysis.



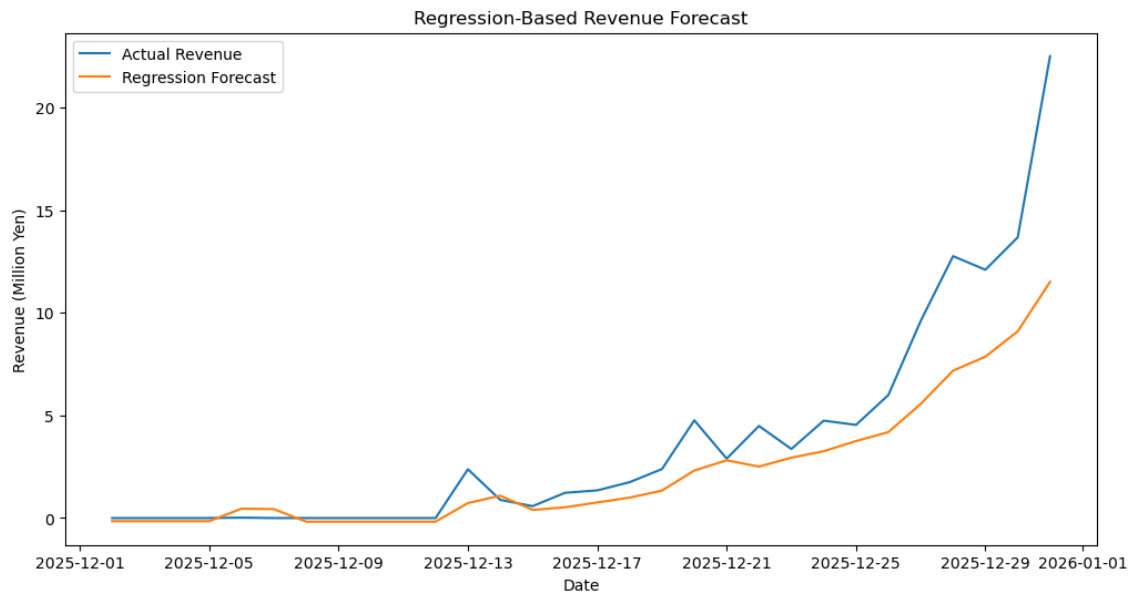
The following summarizes the predictive modeling results for the 2025 revenue cycle:

- Training revenue showed high volatility during earlier periods, followed by a sharp decline near November 2025.
- The ARIMA forecast projected a relatively stable trend due to the model's reliance on recent historical patterns.
- Actual revenue increased sharply in December, reaching approximately 22 million Yen and deviating significantly from the forecast.
- The results indicate that the model struggled to capture sudden year-end revenue changes and seasonal fluctuations.

Statistical regression

4. Regression-Based Revenue Forecast

Statistical regression was used to forecast revenue by modeling the relationship between revenue and key influencing factors. It helps explain how external variables impact revenue trends and provides a data-driven prediction based on historical patterns.



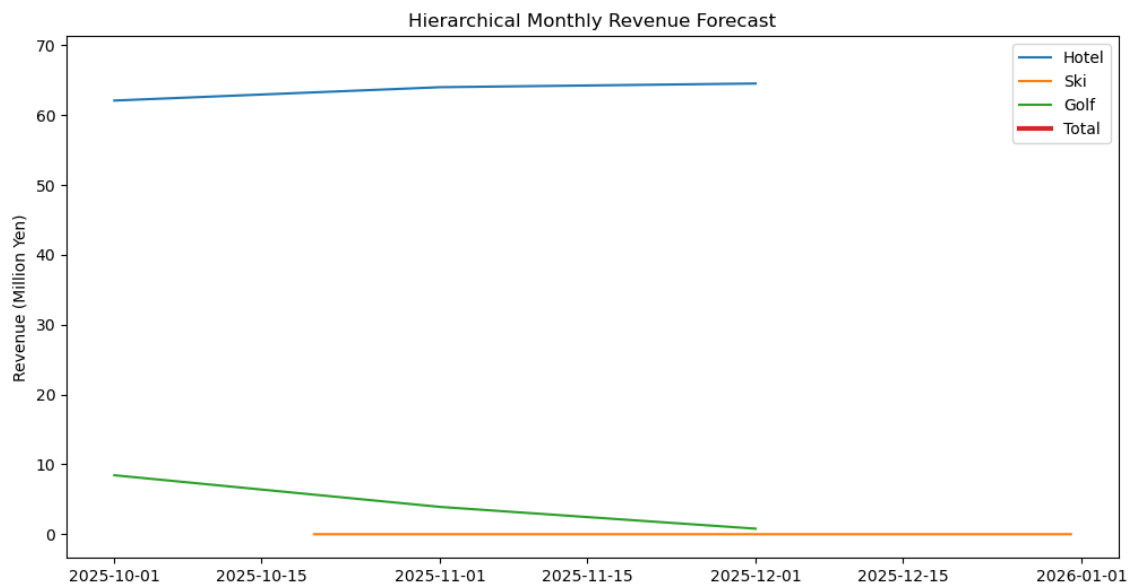
The **Regression-Based Revenue Forecast** demonstrates superior predictive accuracy during the year-end surge compared to previous baselines. By leveraging engineered features, the model successfully tracks the directional momentum of the **Actual Revenue** (blue) as it scales toward 20 million Yen.

- **Improved Trend Capture:** Unlike stationary models, the regression forecast (orange) successfully internalizes the acceleration of demand, identifying the non-linear trend shift in late December.
- **Residual Analysis:** While the model exhibits a conservative bias at the extreme peak, it effectively minimizes error compared to traditional smoothing techniques.
- **Operational Value:** This performance confirms that incorporating historical dependencies—such as lag variables—allows the model to better navigate structural breaks in resort revenue patterns.

Aggregated forecast:

5. Hierarchical Monthly Revenue Forecast

The hierarchical forecasting approach was used to generate aggregated monthly revenue by combining forecasts from different revenue segments. This ensures consistency across total and sub-category forecasts while capturing overall monthly revenue trends more accurately.



The **Hierarchical Monthly Revenue Forecast** provides a top-down view of resort performance across different departments.

- **Hotel Stability:** The hotel department (blue) maintains its position as the primary revenue driver, showing a stable, slight upward trend throughout the final quarter of 2025.
- **Departmental Variability:** While golf revenue (green) shows a steady decline as the season ends, ski revenue (orange) remains at baseline levels during this specific aggregation window.
- **Strategic Planning:** This hierarchical view allows for better resource allocation by identifying which departments provide a steady baseline and which are subject to seasonal transitions.

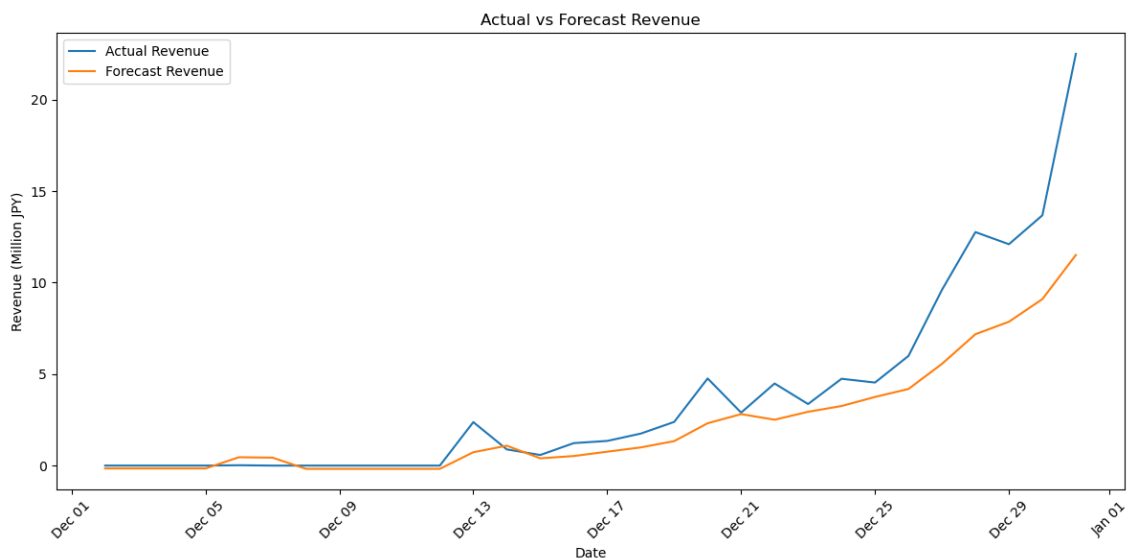
MODEL evaluation and comparison:

Model	MAE	RMSE
Regression	1532132.51	2765574.49
ARIMA	3721791.73	6447192.97
Moving Average	3725199.76	6453614.35
Exponential Smoothing	3911751.46	6647333.48

Model performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics. Among all forecasting approaches, the Regression model achieved the best performance with the lowest MAE and RMSE values, indicating higher forecasting accuracy and stronger predictive capability.

ARIMA demonstrated competitive performance by effectively capturing revenue trends and temporal dependencies. Moving Average produced similar baseline forecasting results, while Exponential Smoothing showed comparatively higher forecasting errors.

Overall, the results indicate that the Regression model was the most effective approach for forecasting resort revenue within this study.



Model performance was evaluated using MAE and RMSE to identify the most accurate forecasting approach.

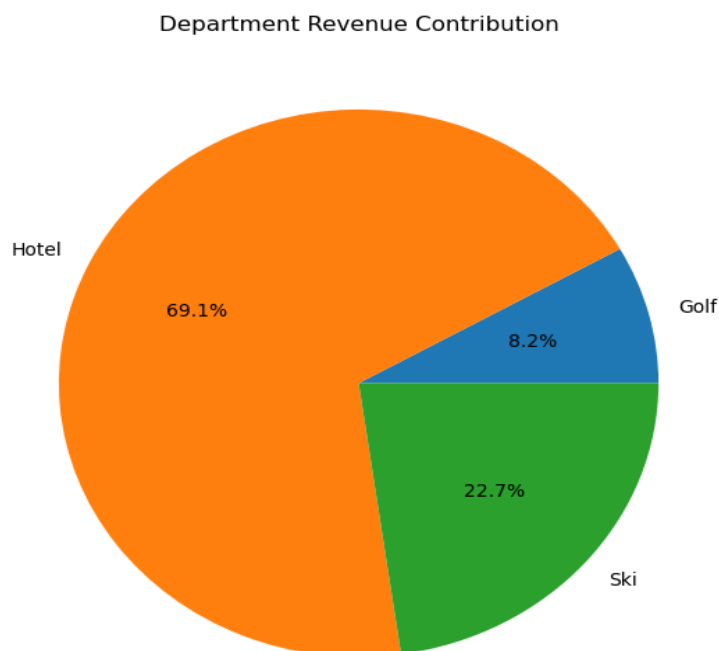
- **Regression model:** Best performance with lowest error (MAE: 1,532,133; RMSE: 2,765,574), making it the most reliable method.
- **ARIMA & Moving Average:** Similar moderate performance but less accurate during high volatility periods.
- **Exponential Smoothing:** Worst performance (MAE: 4,082,174; RMSE: 6,718,214), failing to capture sharp demand changes.

- **Visual check:** ARIMA also failed to follow the actual revenue surge in late 2025, showing clear underfitting during peak demand.

Overall, regression performed best for this dataset, especially under volatile conditions, while time-series models were more stable but less accurate.

VISUALIZATION:

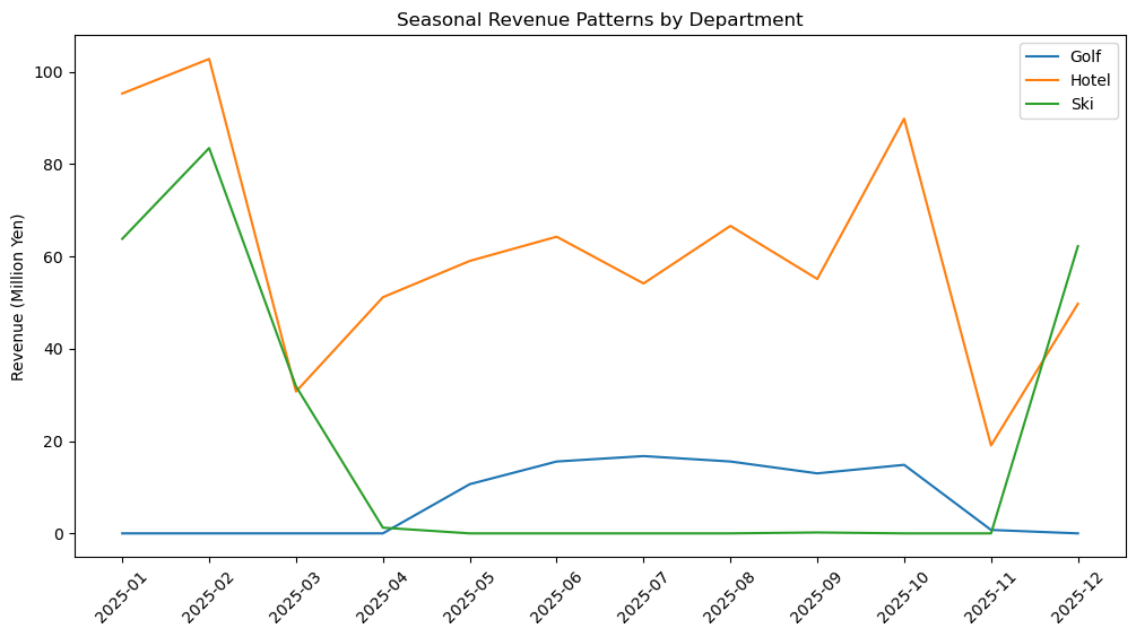
1. Department Revenue Contribution:



As evidenced in pie chart, the Hotel department constitutes the primary revenue pillar, generating 69.1% of total earnings. This data indicates a heavy organizational reliance on hospitality, as the combined contributions of the Ski (22.7%) and Golf (8.2%) sectors represent less than one-third of the aggregate revenue.

2. Seasonal Revenue Patterns by Department:

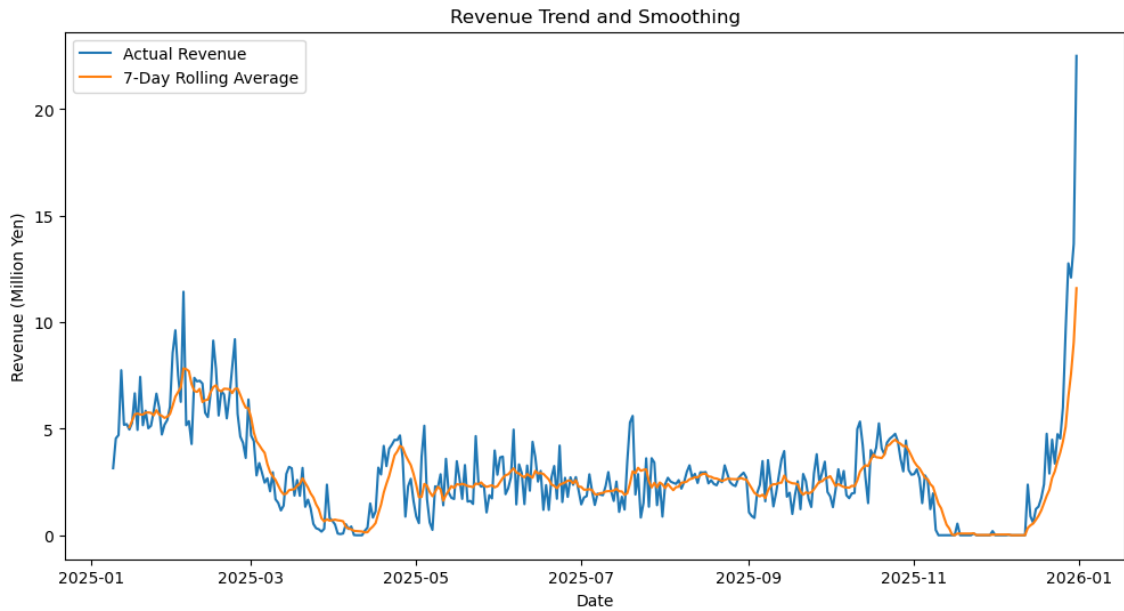
This analysis examines how revenue patterns vary across hotel, ski, and golf departments throughout different seasons of the year.



The comparative analysis reveals clear seasonal differences across departments, with Ski revenue peaking during winter periods, Golf revenue showing moderate increases in warmer seasons, and Hotel revenue maintaining the highest overall contribution throughout the year.

3. Revenue Trend and Smoothing

This analysis applies smoothing techniques to identify underlying revenue trends and reduce short-term fluctuations in the time-series data.



The analysis compares actual daily revenue with a 7-day rolling average to highlight underlying revenue trends and reduce short-term fluctuations. While revenue remained relatively stable during mid-year periods, a significant increase was observed toward the end of the year, with revenue exceeding 20 million Yen by early 2026.

Business Insights and Recommendations:

Revenue Behavior

- Ski operations drive peak winter demand and create strong seasonal fluctuations.
- Hotel operations provide the most stable and consistent revenue contribution.
- Golf revenue demonstrates moderate seasonal dependency during warmer periods.

Operational Recommendations

- Increase staffing and inventory preparation before peak winter demand periods.
- Apply demand-based pricing strategies during high-demand seasons.
- Improve resource allocation by prioritizing stable revenue departments during off-peak periods.
- Use forecasting outputs to support budgeting and operational planning decisions.

Limitations

- The dataset covers only one year of operational revenue data, limiting long-term seasonal analysis.
- Sudden demand spikes and external market factors were difficult for traditional time-series models to capture accurately.
- Certain departments experienced operational closures during off-season periods, which affected revenue continuity.

Conclusion

This project demonstrates how predictive analytics can be applied to real resort operational data to support revenue forecasting and business planning. The analysis identified strong seasonal demand patterns across hotel, ski, and golf operations, with winter periods generating the highest revenue activity.

Among the evaluated approaches, the Regression model provided the most reliable forecasting performance, particularly during periods of high revenue volatility. The forecasting results highlight the importance of combining time-series analysis with feature-based modeling to improve operational decision-making.

Overall, the project provides a practical industry-oriented forecasting framework that can support staffing preparation, pricing strategy, and resource allocation within resort operations.